

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$5(x + 7) = 15(x - 17)(x + 7)$$

What is the sum of the solutions to the given equation?

Correct Answer: 10.33, 31/3

Rationale

The correct answer is $\frac{31}{3}$. Subtracting $5(x + 7)$ from each side of the given equation yields $0 = 15(x - 17)(x + 7) - 5(x + 7)$. Since $5(x + 7)$ is a common factor of each of the terms on the right-hand side of this equation, it can be rewritten as $0 = 5(x + 7)(3(x - 17) - 1)$. This is equivalent to $0 = 5(x + 7)(3x - 51 - 1)$, or $0 = 5(x + 7)(3x - 52)$. Dividing both sides of this equation by 5 yields $0 = (x + 7)(3x - 52)$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $x + 7 = 0$ or $3x - 52 = 0$. Subtracting 7 from both sides of the equation $x + 7 = 0$ yields $x = -7$. Adding 52 to both sides of the equation $3x - 52 = 0$ yields $3x = 52$. Dividing both sides of this equation by 3 yields $x = \frac{52}{3}$. Therefore, the solutions to the given equation are -7 and $\frac{52}{3}$. It follows that the sum of the solutions to the given equation is $-7 + \frac{52}{3}$, which is equivalent to $-\frac{21}{3} + \frac{52}{3}$, or $\frac{31}{3}$. Note that 31/3 and 10.33 are examples of ways to enter a correct answer.